EARS 51 & 151 Mineralogy and Earth Processes
Winter 2026 Syllabus

Lecture: 11 Slot MWF 11:30a-12:35p 414 Fairchild

X-hour: Tuesday 12:15p-1:05p 412a Fairchild

Lab: Wednesday 3:30-5:20pm 412a Fairchild

Instructor: Dr. Thom Chaffee

Office: [redacted]

Email: [redacted]

Office Hours: Mondays 10:30-11:25am in [redacted]

TA: [redacted]

Office: [redacted]

Email: [redacted]

Office Hours: By appointment

Description: Crystallography, mineral chemistry, and physical properties of the principal rock forming minerals, especially the silicates. In the laboratory, technique of hand identification, the petrographic microscope, and X-ray diffraction are mastered.

Learning Goals: By the end of the course, students in EARS 51 and 151 will be able to:

-Use a range of physical, chemical, and optical properties to identify different minerals in hand sample and thin section. Recognize and distinguish the major rock-forming minerals.

-Analyze and interpret the chemical and structural makeup of many different mineral groups, and how these properties affect observable physical and optical mineral features.

-Describe formation processes that lead to the creation of different mineral species and different occurrences within a mineral species; and consequently, use observations and measurements of mineral samples to infer what processes and conditions they have experienced.

Additionally, students in EARS 151 will be able to:
-Consult with research literature to investigate a mineral species and summarize their findings in written form.

Textbook: The main text for the course, covering both lectures and labs, is
Nesse WD. *Introduction to Mineralogy*. Fourth edition. Oxford University Press; 2024.
A widely used text that covers the lecture material. The 2nd and 3rd editions are also good resources (1st edition is somewhat outdated)

Other good sources of information:

Wenk HR, Bulakh AG. *Minerals: Their Constitution and Origin*. Second edition. Cambridge University Press; 2016.
This textbook is available as an ebook through the Dartmouth library with simultaneous access for up to 3 users.

Nesse WD. *Introduction to Optical Mineralogy*. 2nd ed. Oxford University Press; 1991.
A great reference for the laboratory material

Other written reference material is available in Fairchild 412a for use during labs.

Mindat.org has a wealth of useful mineral information.

Wikipedia is not a definitive source but often provides good overviews of mineral species, with citations to relevant publications.

Pre-requisites:

EARS 51 is designed for EARS majors and minors in their junior and senior years. **EARS 40** covers some introductory topics in mineralogy and is a requirement for taking the course. **CHEM 5** or equivalent is also required: the study of mineralogy builds directly on concepts from chemistry. EARS majors who do not meet these prerequisites may contact the instructor to discuss waiving them. Students outside the EARS department who have completed courses in chemistry and/or materials science are encouraged to contact the instructor.

EARS 151 is intended for graduate students who have previously completed a general chemistry course. If you have previously taken mineralogy, this course may provide an opportunity for an in-depth review of topics.

Students not in the EARS department who are interested in EARS 51 / 151 are encouraged to contact the instructor prior to enrollment; completed coursework in chemistry or materials science is recommended.

Course Activities:

Lecture: Lectures include delivery of material from the instructor in conjunction with interactive student activity components. **Participation** is important in this course and is graded based on weekly submitted in-class work; please let the instructor know in advance if you are unable to attend a lecture and a suitable make-up activity will be provided. See the course schedule below for an overview of lecture topics.

Laboratory: Weekly lab exercises will be supervised by the instruction team, and graded. Most weeks, the laboratory period will be spent in Fairchild 412a using optical petrographic microscopes to look at thin sections. Near the end of the quarter, we will use modern mineral identification methods including electron microscopy and spectroscopy (SEM + EDS) and X-Ray Diffraction (XRD).

Field Trip: There will be an optional one-day field trip during the quarter. Details TBD.

Exams: There will be an in-class midterm (Monday, Feb 9th) and a take-home final during the exam period (March 13-17), both based on the lectures. The written exams will be mostly short response; the final will mostly cover the 2nd half of the course material. There will also be an in-person laboratory exam during the final exam period (Saturday, March 14, 11:30 AM), involving identification of minerals in hand sample and thin section.

151 Only: Final Paper Students enrolled in 151 will complete a research review paper of circa 2500 words on a mineralogy topic of their choosing with at least 15 sources (10 of them must be peer-reviewed literature). This writing project is intended to help graduate students with their research and in developing their scientific communication skills. Suggested topic: choose a mineral or mineral family relevant to your research, do a search of peer-reviewed literature studying that mineral within your subfield, and write a paper summarizing the approaches and findings of 10 sources and providing some of your own commentary on the work. See the course schedule below for more details.

Course Policies:

Canvas: The instruction team will use Canvas to post grades, share materials, and make course announcements. Some assignments will be turned in through Canvas, others as paper copies. You may contact the instructors through Canvas or by direct email (preferred).

Grading: Every effort will be made to return graded material within one week of its due date, and not more than two weeks. The final grade for EARS 51 will consist of:

Weekly Lab Exercises 45%; In-Class Activities 10%; Midterm Exam 15%; Final Exam 30%

The final grade for EARS 151 will consist of:

Weekly Lab Exercises 22.5%; In-Class Activities 10%; Midterm Exam 15%; Final Exam 30%; Final Paper 22.5%

Late Policy & Extensions: Extensions of one week are granted up to 24 hours prior to the due date for assignments (not exams) with no questions asked. Late assignments without an extension are penalized at 10% per day.

Exams: In-class exams will be closed-notes, and no collaboration. They will consist of conceptual questions as well as identification of mineral specimens. The midterm will have an opportunity to correct wrong answers for 50% of the lost points. The take-home final will be open notes, but no collaboration; it will consist of conceptual questions.

Honor, Collaboration, Plagiarism, and AI:

All students are expected to follow the Academic Honor Principle as well as the Guarini Honor Policy and/or Arts and Sciences Honor Policy.

Learning is best accomplished communally, and I encourage students to collaborate during lab exercises and lecture activities via discussion; however, you must prepare and submit your own assignments (see below). Collaboration is not permitted on exams. Collaboration on the final paper for 151 via peer editing is permitted with prior instructor approval.

All work turned in to the instruction team for grading is to be original and solely authored and edited by you as an individual unless otherwise identified and cited or otherwise explicitly permitted in advance by the instructor in writing. Missing or improper attribution is a form of academic dishonesty and subject to the Academic Honor Principle and school Academic Honor Policies.

In alignment with the guidelines on using generative AI the use of Large Language Model (LLM) and/or what are commonly called Artificial Intelligence (AI) or generative AI tools is

not permitted in this course. The instruction team will not use them in teaching, and expect you to reciprocate in your learning.

Accessibility and Accommodations:

Students requesting disability-related accommodations and services for this course are required to register with Student Accessibility Services (SAS; Apply for Services webpage; student.accessibility.services@dartmouth.edu; 1-603-646-9900) and to request that an accommodation email be sent to me in advance of the need for an accommodation. Then, students should schedule a follow-up meeting with me to determine relevant details such as what role SAS or its Testing Center may play in accommodation implementation. This process works best for everyone when completed as early in the quarter as possible. If students have questions about whether they are eligible for accommodations or have concerns about the implementation of their accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Religious Observances:

Dartmouth and your instruction team have a deep commitment to support students' religious observances and diverse faith practices. Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible—before the end of the second week of the term at the latest—to discuss appropriate course adjustments.

Mental health and wellness:

The academic environment is challenging, our terms are intensive, and classes are not the only demanding part of your life. There are a number of resources available to you on campus to support your wellness, including: the Counseling Center which allows you to book triage appointments online, the Student Wellness Center which offers wellness check-ins, and your undergraduate dean. The student-led Dartmouth Student Mental Health Union and their peer support program may be helpful if you would like to speak to a trained fellow student support listener. If you need immediate assistance, please contact the counselor on-call at (603) 646-9442 at any time. Please make me aware of anything that will hinder your success in this course.

Class Climate and Inclusivity

The EARS Department is dedicated to creating a positive, inclusive work and learning environment that embraces diversity in all forms and rejects any form of hostile workplace, discrimination, or bullying. Respect for and civility toward other students and the instructors is expected.

Course Schedule and Topics:

This section is an outline and may change during the quarter based on the interests of students, the rate of completion of material, and any unexpected circumstances. An updated schedule will be maintained on Canvas. There is one built-in buffer day in case of winter weather or if some material needs an extra day of lecture.

Most lectures have associated suggested reading in *Wenk & Bulakh (W&B*) and in *Nesse* 3rd edition. Some chapters are split between multiple lectures: read the parts of the chapter relevant to the lecture topic.

The X-hour will be used weekly as an optional additional lab period and instructor office hour. This is a good time to finish the previous week's lab assignment.

All labs include a written assignment due by the beginning of the following lab.

Week 1 (5 Jan):

M) Introduction to the course and minerals.

W) Mineral properties in thin section *Reading: W&B Ch. 14.1-14.3, 14.6 (Ch. 13 also covers optical physics) / Nesse Ch. 7*

Lab 1 (7 Jan): Introduction to the optical petrographic microscope and thin section properties.

F) Mineral properties in hand sample *Reading: W&B Ch. 6 & Ch. 12 / Nesse Ch. 6*

Week 2 (12 Jan):

M) Crystal Chemistry *Reading: W&B Ch. 4+5 (Ch. 2 for chemistry review) / Nesse Ch. 3*

W) Crystallography and Symmetry *Reading: W&B Ch. 7+8 / Nesse Ch. 2*

Lab 2 (14 Jan): Crystal structures and symmetry. Lab 1 due at start of lab period.

F) Crystal structures and defects *Reading: W&B Ch. 9+10 / Nesse Ch. 4*

Week 3 (19 Jan):

M) NO CLASS – MLK DAY

W) Advanced optical mineralogy *Reading: W&B Ch. 14 (§14.4-14.7) / Nesse Ch. 7 (p. 165-189)*

Lab 3 (21 Jan) Identifying common minerals in thin section.

F) Crystal Growth – post-crystallization, geology, thermodynamics *Reading: W&B Ch. 18+19 / Nesse Ch. 5*

Week 4 (26 Jan):

M) Introduction to silicates *Reading: W&B Ch. 21 / Nesse Ch. 11*

W) Orthosilicates and disilicates *Reading: W&B Ch. 28 / Nesse Ch. 15+16*

Lab 4 (28 Jan) Silicate minerals I

F) Sheet silicates *Reading: W&B Ch. 29 / Nesse Ch. 13*

Week 5 (2 Feb):

M) Chain silicates *Reading: W&B Ch. 30 / Nesse Ch. 14*

Midterm review during X-hour Week 5 (Feb 3rd)

W) Framework silicates *Reading: W&B Ch. 31 / Nesse Ch. 12*

Lab 5 (4 Feb) Silicate minerals II

F) Ring silicates (cyclosilicates) *Reading: W&B Ch. 28 / Nesse Ch. 15*

Week 6 (9 Feb):

M) Midterm

W) Halides *Reading: W&B Ch. 23 / Nesse Ch. 18*

Lab 6 (11 Feb) Halides & Sulfides

F) Sulfides *Reading: W&B Ch. 26 / Nesse Ch. 19*

Week 7 (16 Feb):

M) Midterm review and corrections

W) Phosphates & Sulfates *Reading: W&B Ch. 25 / Nesse Ch. 17*

Lab 7 (18 Feb) Phosphates & Sulfates

F) Pegmatites (TA lecture) *Reading: N/A*

Sat. 21 Feb *- Field trip to museum (pending their confirmation)*

Week 8 (23 Feb):

M) Electron spectroscopy *Reading: W&B Ch. 16 / Nesse Ch. 9+10*

W) X-ray crystallography *Reading: W&B Ch. 11 / Nesse Ch. 8+10*

Lab 8 (25 Feb) Energy dispersive spectroscopy and X-ray diffraction (Fairchild 219)

F) Oxides and Hydroxides *Reading: W&B Ch. 27 / Nesse Ch. 18*

Week 9 (2 Mar): -

M) Carbonates *Reading: W&B Ch. 24 / Nesse Ch. 17*

W) Machine learning and mineral identification *Reading: Paper posted in Canvas*

Lab 9 (4 Mar) Carbonates, Oxides, Hydroxides, and Miscellaneous

F) Buffer day (in case we are behind on lectures)

Week 10 (9 Mar):

M) Extraterrestrial minerals. Final class. *Reading: W&B Ch. 37*

Final review during X-hour Week 10. (Mar 10th)

Finals

Final Exam: Take-home exam available March 13th, due March 17th

Saturday, March 14, 11:30 AM In-person Exam (Fairchild 412a)

151 Students: Final Paper Schedule

Friday, Jan 16 (Week 2): Topics due.

Friday, Jan 30 (Week 4): Completed outline and 3 annotated sources due.

Friday, Feb 6 (Week 5): Instructor feedback on outline and annotations provided.

Friday, Feb 13 (Week 6): First draft due.

Friday, Feb 20 (Week 7): Instructor feedback on first draft provided.

Friday, Mar 6 (Week 9): Final drafts due.